A higher ratio of beans to white rice is associated with lower cardiometabolic risk factors in Costa Rican adults

Background: A high intake of white rice is associated with the metabolic syndrome and type 2 diabetes. Costa Ricans follow a staple dietary pattern that includes white rice and beans, yet the combined role of these foods on cardiometabolic risk factors has not been studied. Objective: We aimed to determine the association between intake of white rice and beans and the metabolic syndrome and its components in Costa Rican adults (n = 1879) without diabetes. Design: Multivariate-adjusted means were calculated for components of the metabolic syndrome by daily servings of white rice and beans (<1, 1, or >1) and by the ratio of beans to white rice. The OR for the metabolic syndrome was calculated by substituting one serving of beans for one serving of white rice. Results: An increase in daily servings of white rice was positively associated with systolic blood pressure (BP), triglycerides, and fasting glucose and inversely associated with HDL cholesterol (P-trend <0.01 for all). An increase in servings of beans was inversely associated with diastolic BP (P = 0.049). Significant trends for higher HDL cholesterol and lower BP and triglycerides were observed for 1:3, 1:2, 1:1, and 2:1 ratios of beans to white rice. Substituting one serving of beans for one serving of white rice was associated with a 35% (95% CI: 15%, 50%) lower risk of the metabolic syndrome. Conclusion: Increasing the ratio of beans to white rice, or limiting the intake of white rice by substituting beans, may lower cardiometabolic risk factors.